



DEPARTMENT OF MARINE TECHNOLOGY

TTK4190 - GUIDANCE, NAVIGATION AND CONTROL OF MARINE
CRAFT, AIRCRAFT AND DRONES

Assignment 1 - Kinematics and Kinetics

Authors:

Audun Bagøien Hustad
Bendik Letting Skari
Fredrik Siem Taklo
William Hammer

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1.1 Equilibrium & Linearization

1.1.1 Equilibrium point

The attitude kinematics in quaternion form are

$$\dot{q} = T_q(q) \omega, \quad q = \begin{bmatrix} \eta \\ \epsilon^\top \end{bmatrix}, \quad (1)$$

with $\eta \in \mathbb{R}$ the scalar part and $\epsilon \in \mathbb{R}^3$ the vector part of the quaternion. The rotational dynamics are

$$I_g \dot{\omega} - S(I_g \omega) \omega = \tau, \quad (2)$$

where $I_g = \frac{mL^2}{6} I_3$ for the cube satellite.

The equilibrium corresponding to

$$q = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \tau = 0,$$

is

$$x_0 = \begin{bmatrix} \epsilon \\ \omega \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

1.1.2 Linearization

Around $x = x_0$, the dynamics can be approximated as

$$\dot{\epsilon} = \frac{1}{2} \omega, \quad (3)$$

$$\dot{\omega} = I_g^{-1} \tau, \quad I_g = \frac{mL^2}{6}. \quad (4)$$

Thus, the linearized state-space system

$$\dot{x} = Ax + B\tau$$

has matrices

$$A = \begin{bmatrix} 0_{3 \times 3} & \frac{1}{2} I_3 \\ 0_{3 \times 3} & 0_{3 \times 3} \end{bmatrix}, \quad B = \begin{bmatrix} 0_{3 \times 3} \\ I_g^{-1} I_3 \end{bmatrix}.$$

1.2 PD Attitude Control

The control law for attitude is defined as follows.

$$\tau = -\mathbf{K}_d \omega - k_p \epsilon$$

Where the given controller gains is set to

$$k_p = 5 \times 10^{-5}, \quad k_d = 9 \times 10^{-4}.$$

$$K = [-k_p I_3 \quad -\mathbf{K}_d].$$

The closed-loop state matrix is

$$A_{cl} = A - BK,$$

with eigenvalues obtained from the MATLAB implementation in Appendix A.

$$\lambda = \{-0.0270 \pm 0.0278i, -0.0270 \pm 0.0278i, -0.0270 \pm 0.0278i\}.$$

The eigenvalues are shown to have a real negative part, which is desirable for a stable system. The imaginary part also implies there is a slight oscillation in the system before converging to the, as the system is underdamped. For the satellite, a more damped system would be more preferred, as oscillations can lead to unnecessary waste of energy. Therefore, it would be desirable to have strict negative poles with only the real value.

1.3 Euler-Angle Simulation

To simulate the Euler-angles, the provided attitude.m file was used. Some changes had to be made to match our system. The first to change are setting the initial conditions to:

$$[\phi(0) = -5^\circ, \theta(0) = 10^\circ, \psi(0) = -20^\circ]$$

Then the control law had to be changed to:

$$\tau = -k_d \cdot \dot{q} - k_p \cdot q;$$

With $k_p = 5 \times 10^{-5}$ and $k_d = 9 \times 10^{-4}$.

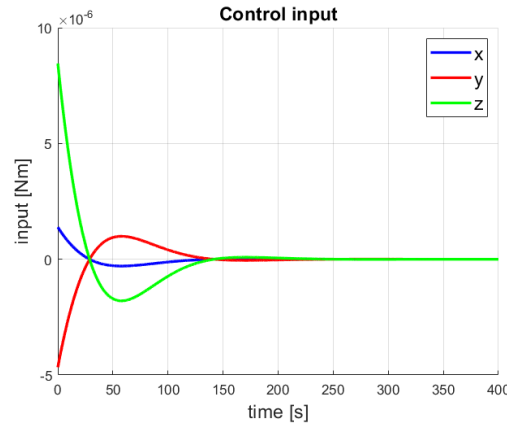


Figure 1: Plot showing the control inputs.

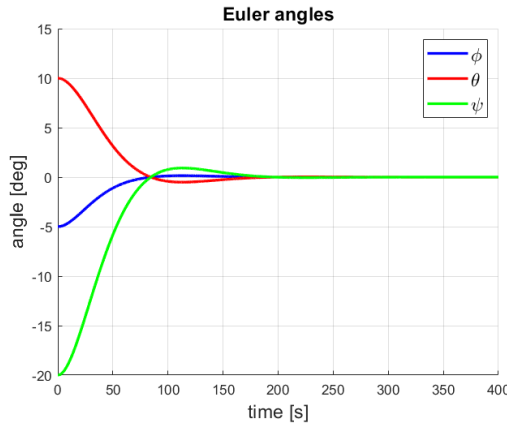


Figure 2: Plot showing the resulting Euler angles.

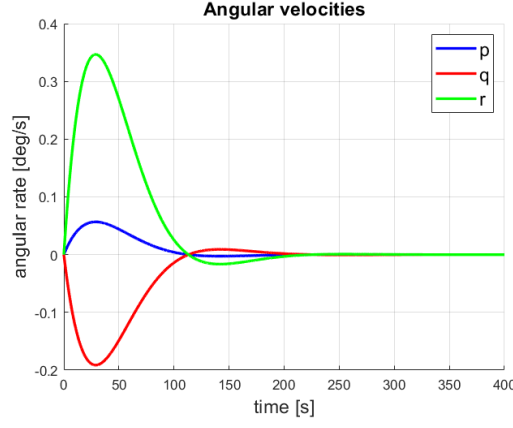


Figure 3: Plot showing the angular velocities.

We can see in Figure 10 and 11 that the response behaves like described in task 1.2. There is a big overshoot, but it eventually stabilizes at 0. This is because of the imaginary poles that results in some oscillations.

To follow a nonzero constant reference signal, one can change the control law to include quaternion error. This can be done by replacing ϵ with ϵ_{error} in the control law.

1.4 Quaternion Error

The quaternion error is defined as:

$$\tilde{q} = \bar{q}_d \otimes q$$

Where $\bar{q}_d$ is the conjugate of the reference quaternion:

$$\bar{q}_d = \begin{bmatrix} \eta_d \\ -\epsilon_d \end{bmatrix}$$

Computing the quaternion product gives us:

$$\tilde{q} = \bar{q}_d \otimes q = \begin{bmatrix} \eta_d \eta + \epsilon_d^T \epsilon \\ \eta_d \epsilon - \eta \epsilon_d - S(\epsilon_d) \epsilon \end{bmatrix}$$

Expanding this to component form, looking at only $\tilde{\epsilon}_i$, we get:

$$\begin{aligned} \tilde{\epsilon}_1 &= \eta_d \epsilon_1 - \eta \epsilon_{d1} - (\epsilon_{d2} \epsilon_3 - \epsilon_{d3} \epsilon_2), \\ \tilde{\epsilon}_2 &= \eta_d \epsilon_2 - \eta \epsilon_{d2} - (\epsilon_{d3} \epsilon_1 - \epsilon_{d1} \epsilon_3), \\ \tilde{\epsilon}_3 &= \eta_d \epsilon_3 - \eta \epsilon_{d3} - (\epsilon_{d1} \epsilon_2 - \epsilon_{d2} \epsilon_1). \end{aligned}$$

After convergence $\tilde{q}$ should be close to zero, as this represents the error. That also means $\mathbf{q} = \mathbf{q}_d$. The attitude is the same as the desired attitude when there is no error.

1.5 Tracking q_d

A few more changes are needed to the code in order to follow a quaternion reference. The reference quaternion can be constructed using the euler2q function from the MMS toolbox:

```
qd = euler2q(0, deg2rad(15)*cos(0.1*t(i)), deg2rad(10)*sin(0.05*t(i)));
```

The error gets calculated using the quaternion product derived in Problem 1.4:

```
eps_d = qd(2:4);  
eta_d = qd(1);  
eps = q(2:4);  
eta = q(1);  
  
eps_err = eta_d*eps - eta*eps_d - cross(eps_d, eps);
```

The control law is also adjusted to include the quaternion error:

```
tau = -kd*eye(3)*w - kp*eps_err;
```

Figure 4 shows that the satellite are not able to follow the given reference signal. It is oscillating instead of converging. This was not the expected response. Increasing the gains gives a somewhat better response. However, its still oscillating too much. Further increasing the gains results in an unstable system.

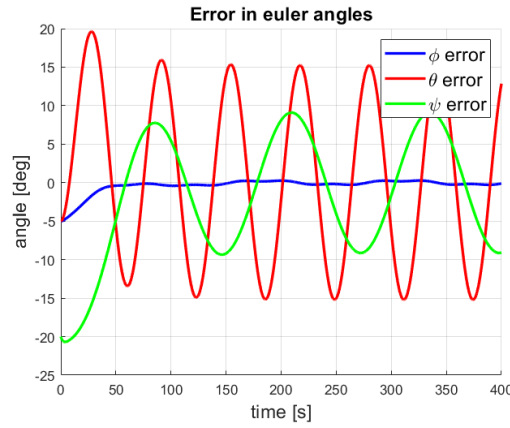


Figure 4: Plot showing the Euler error.

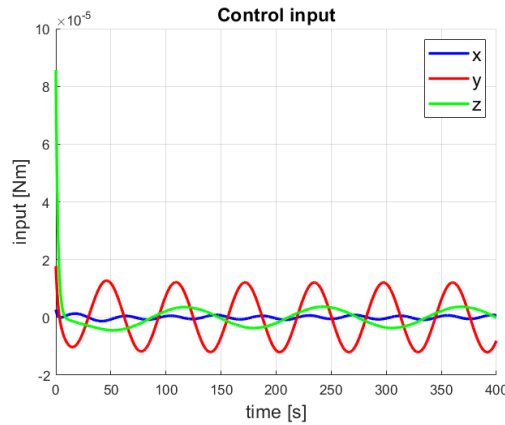


Figure 5: Plot showing the control inputs.

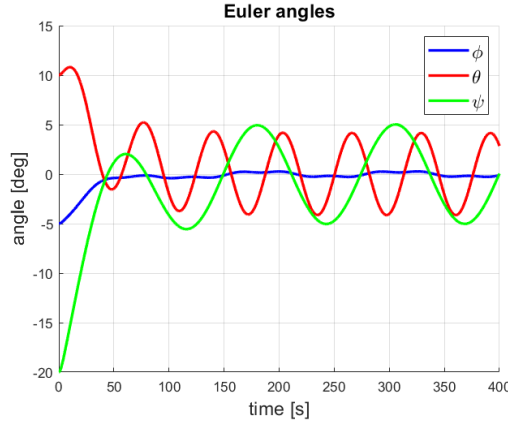


Figure 6: Plot showing the resulting Euler angles.

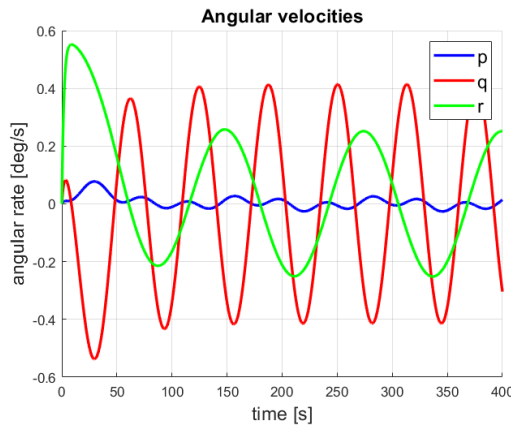


Figure 7: Plot showing the angular velocities.

1.6 Angular-Rate Error PD

The controller in Problem 1.5 tried to follow the quaternion reference and at the same time keep the angular velocity zero. This leads to issues because the reference is constantly moving and are not zero. By following the angular velocity error instead of a zero reference, one should get a better response. This can be done with the control law:

$$\tau = -\mathbf{K}_d \tilde{\omega} - k_p \tilde{\varepsilon}$$

To calculate the angular velocity error $\tilde{\omega}$, one can first calculate the Euler angle rate reference $\dot{\Theta}_d$ by derivation of the given Euler angle reference signal and convert to radians to get:

$$\dot{\Theta}_d(t) = \begin{bmatrix} 0 \\ -0.0262 \sin(0.1t) \\ 0.00873 \cos(0.05t) \end{bmatrix}$$

The reference signal for the Euler angle rate can be calculated by transforming $\dot{\Theta}_d$ using the transformation $T^{-1}(\Theta_{nb})$ from the lecture notes. (Fossen 2011):

$$T^{-1}(\Theta_d) = \begin{bmatrix} 1 & 0 & -\sin \theta \\ 0 & \cos \phi & \cos \theta \sin \phi \\ 0 & -\sin \phi & \cos \theta \cos \phi \end{bmatrix}$$

$$\omega_d = T^{-1}(\Theta_d) \dot{\Theta}_d$$

From here, the angular velocity error can be calculated using.

$$\tilde{\omega} = \omega - \omega_d$$

Simulating this gives us a better response than in Problem 1.5. Figure 8 shows that there are still oscillations, but with a much lower amplitude than in Figure 4.

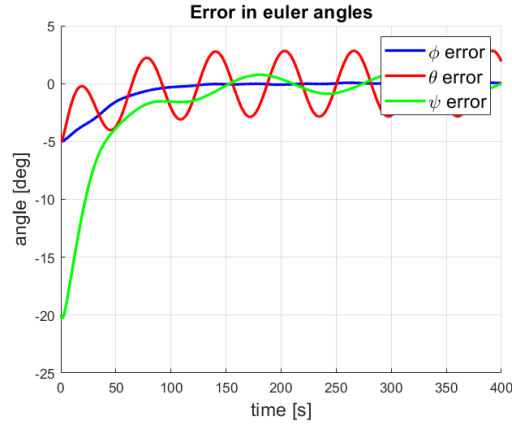


Figure 8: Plot showing the Euler error, now tracking both reference angle and reference angle velocity.

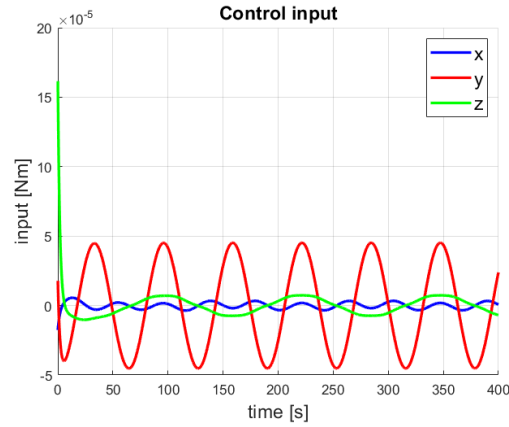


Figure 9: Plot showing the control inputs.

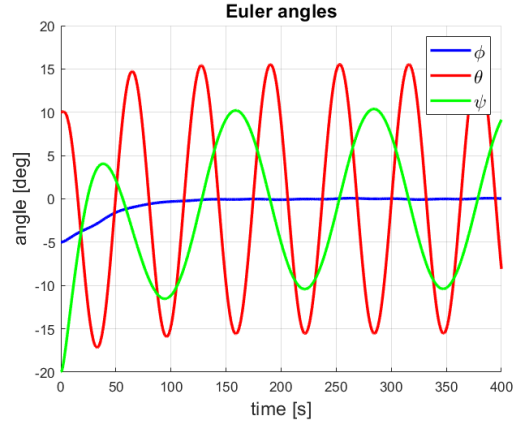


Figure 10: Plot showing the resulting Euler angles.

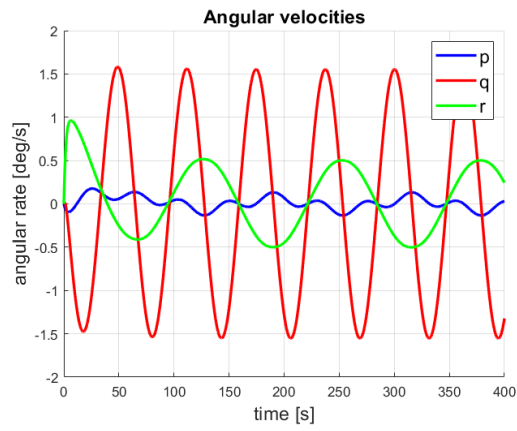


Figure 11: Plot showing the angular velocities.

1.7 Lyapunov Stability

The closed loop system is analysed using Lyapunov stability. The candidate Lyapunov function is defined in the given task.

$$V := \frac{1}{2} \tilde{\omega}^\top \mathbf{I}_g \tilde{\omega} + 2k_p(1 - \tilde{\eta}) \quad (5)$$

1.7.1 Derivative of the Lyapunov function

Performing the derivative of Equation 5 yields

$$\begin{aligned}
\dot{V} &= \frac{\partial V}{\partial x} \frac{\partial x}{\partial t} \\
&= \dot{\omega}^\top I_g \dot{\omega} - 2k_p \dot{\eta} \\
&= \omega^\top I_g \dot{\omega} - 2k_p \left(-\frac{1}{2} \epsilon^\top \omega\right) \\
&= \omega^\top I_g \dot{\omega} + k_p \epsilon^\top \omega \\
&= \omega^\top I_g (I_g^{-1} (\tau - S(I_g \omega))) + k_p \epsilon^\top \omega \\
&= \omega^\top \tau + k_p \epsilon^\top \omega \\
&= \omega^\top (-k_d \omega - k_p \epsilon) + k_p \epsilon^\top \omega \\
&= -k_d \omega^\top \omega - k_p \omega^\top \epsilon + k_p \epsilon^\top \omega \\
&= -k_d \omega^\top \omega - k_p \omega^\top \epsilon + k_p (\epsilon^\top \omega)^\top \\
&= -k_d \omega^\top \omega.
\end{aligned}$$

Thus,

$$\dot{V} = -k_d \omega^\top \omega < 0, \quad \tilde{\omega} \neq 0$$

where $k_d > 0$ is positive definite, and the error ϵ cancels out in $\dot{V}$.

1.7.2 Stability analysis of the Lyapunov function

The candidate Lyapunov function is positive definite for $\tilde{\omega} \neq 0$, as

$$\mathbf{I}_g > 0$$

and

$$\tilde{\eta} = \pm 1$$

The derivative of the Lyapunov function is negative definite for $\tilde{\omega} \neq 0$. Therefore, the LaSalle's invariance principle is utilized to check if the system is asymptotically stable.¹ The invariant set where $\dot{V} = 0$ is defined to be

$$\Omega := \left\{ \tilde{\omega} \in \mathbb{R}^3, \tilde{\eta} \in \mathbb{R} \mid \dot{V} = 0 \right\}.$$

From the condition $\dot{V} = 0$ we obtain

$$\Omega = \{ \tilde{\omega} = 0, \tilde{\eta} \in \mathbb{R} \}.$$

Furthermore,

$$\dot{\tilde{\eta}} = -\frac{1}{2} \tilde{\epsilon}^\top \cdot 0 = 0, \quad I_g \dot{\omega} = \tau = -k_d 0 - k_p \tilde{\epsilon} = 0.$$

Thus, if $\tilde{\epsilon} = 0$ it follows that

$$\tilde{\eta} = \sqrt{1 - \tilde{\epsilon}^\top \tilde{\epsilon}} = \pm 1,$$

and the invariant set reduces to

$$M = \{ \tilde{\omega} = 0, \tilde{\eta} = \pm 1 \}.$$

Since M is the largest invariant set in Ω , the system is asymptotically stable. However, due to the existence of multiple equilibrium points, the system is not globally asymptotically stable (GAS).

¹Khalil 2015.

Bibliography

- Fossen, Thor I. (2011). *Handbook of Marine Craft Hydrodynamics and Motion Control*. 1st. Chichester, UK: John Wiley & Sons Ltd. ISBN: 9781119991496.
- Khalil, Hassan K. (2015). *Nonlinear Control*. 1st. Harlow, England: Pearson Education Limited. ISBN: 9781292060507.

Appendix

A Code for 1.2

```
% Model parameters
m = 10;                % Mass
R = 0.1 / sqrt(6);    % Radius of gyration
Ib = m * R^2 * eye(3); % Inertia matrix in CD
I_inv = inv(Ib);

Kp = 5e-5;
Kd = 9e-4*eye(3);

% Defining linearized system matrix and control input matrix
A = [zeros(3,3), 0.5*eye(3);
     zeros(3,3), zeros(3,3)];

B = [zeros(3,3);
     I_inv*eye(3)];

% Control law
tau = [Kp*eye(3), Kd];

A_cl = A-B*tau;

eig_Acl = eig(A_cl);

display(eig_Acl);
```